

PUNJAB RAINWATER HARVESTING

Sector: Agriculture

Hazard addressed: Droughts

Targeted assets: Buffalo, Cattle, Goat, Sheep



Description: This measure involves development of water harvesting systems for livestock-dependent communities to capture and store rainwater and runoff for livestock consumption during dry periods. The intervention improves water availability, reduces pressure on groundwater resources, supports livestock productivity, and enhances resilience of pastoral and agro-pastoral systems against drought and climate variability.

Priority Districts: Bahawalpur, Dera Ghazi Khan, Rajanpur, Bhakkar, Mianwali, Khushab, Chakwal, Attock, Bahawalnagar, Jhang, Jhelum, Lahore, Layyah, Lodhran, Mandi Bahaudding, Muzaffargarh, Okara, Rahim Yar Khan, Rawalpindi, Sargodha.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Water harvesting unit / storage structure.

Scale assumptions: Applicable in drought-prone livestock production areas and rural settlements vulnerable to seasonal water shortages.

Key design parameters: Rainwater/runoff collection systems, storage ponds or tanks, seepage control measures, livestock access points, filtration systems, and water conveyance structures.

Time horizon: Start implementation: 2027 onwards; systems expected to remain functional over long-term periods with proper maintenance.

COST ASSUMPTIONS

Unit cost: PKR 126,250 per unit with 96 ft³ storage capacity (~ USD 451 per unit).

Capital Expenditure (CAPEX): USD 29.5 million per year (2027 – 2035). Includes excavation/construction of storage structures, runoff collection systems, filtration units, and associated infrastructure.

Operational Expenditure (OPEX): USD 2.7 million per year (2036 – 2050). Includes cleaning, repair, desilting, and routine maintenance.

IMPACT

Risk reduction level: Moderate to High.

Time to effectiveness: Immediate to short-term following installation and operationalization.

Expected impact: Improved water availability for livestock during drought periods, reduced dependence on unreliable water sources, enhanced livestock productivity, and strengthened resilience of livestock systems against climate stress.

CO-BENEFITS & TRADE-OFFS

Environmental: Improved rainwater utilization, reduced groundwater abstraction, and enhanced local water conservation.

Social: Improved water access and reduced vulnerability of pastoral and livestock-dependent communities.

Economic: Reduced livestock productivity losses during droughts and improved livelihood stability.

Potential risks: Limited storage capacity during prolonged droughts, maintenance challenges, and contamination risks if systems are poorly managed.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Community participation, technical expertise, financing support, and awareness regarding water harvesting systems.

Required institutions: Livestock Department Punjab, Irrigation Department Punjab, Local Government Department, Rural Development Agencies, and development partners.

Barriers: Initial installation costs, limited technical capacity, maintenance challenges, and weak institutional support systems.

BENEFICIARIES

Primary beneficiaries: Livestock farmers, pastoral communities, rural households, and water-scarce settlements.

Distributional aspects: Particularly beneficial for drought-prone and underserved livestock-dependent communities with limited access to reliable water supplies.